Oliver Chang

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Berkeley, CA | oliverhchang@berkeley.edu | (408) 203-9356 | Earliest Starting Date: May 18, 2026

EDUCATION

University of California, Berkeley — Expected Graduation: May 2028 | GPA: 3.98
B.S. Mechanical Engineering, Tau Beta Phi

RELEVANT EXPERIENCE

Formula Electric at Berkeley — Sept 2024 - Present
Chassis - Mechanical Engineer
- Responsible for 2025 vehicle's steel spaceframe, executing over 30 FEA iterations in ANSYS to optimize tube diameters and node locations, resulting in a 25% improvement in the stiffness-to-weight ratio compared to previous year.
- Created vehicle dynamic models that account for chassis flexibility to understand impact on suspension compliance, quantifying the impact of torsional stiffness on LLTD/steering ratio/tunability/compliance.
- Designed and fabricated torsional stiffness jig to physically validate the FEA model, achieving a margin of error <15% while refining simulation models for improved accuracy.
- Performed 3-point bend tests on welded samples using an Instron machine per ASTM E190 to characterize weld strength and enhance simulation fidelity as well as liquid penetration testing per ASTM E1417.

Theoretical & Applied Fluid Dynamics Laboratory (TAFLAB) — May 2025 - Present
Undergraduate Researcher - Responsible Engineer
- Engineered a spherical electromagnetic energy harvester utilizing a spherical magnet and base of copper coils
- Used ANSYS Maxwell to parametrically sweep and optimize magnet and coil topologies, validated to 6.3% error.
- Applied DFM principles for engineering a modular 3D-printed housing to secure a 6kg high-inertia magnet array while maintaining a precision 3mm air gap under dynamic oscillating loads
- Integrated rectifying circuitry to convert AC signals into regulated DC voltage for battery charging.

UC Berkeley's Space Technology and Rocketry (STAR) — Sept 2024 - May 2025
Propulsion - Manufacturing Engineer
- L2 Rocket: Designed layout in OpenRocket, fabricated airframe, and integrated custom-soldered avionics to record flight data, culminating in a successful launch and recovery from 2,944 ft.
- Liquid Engine: Applied DFM to manufacture 9 custom injector and combustion chamber components; used GD&T to ensure strict precision and tolerances for CNC/mill/lathe for successful static hot fire.

Fremont High Robotics FRC Team 3501: Firebots — 2020 - 2024
VP of Mechanical Design
- Led a 6-person CAD team using Onshape, iterating a competition robot that placed in top 15% of 3,000+ teams worldwide.
- Managed all parts procurement (BOM), technical drawing approval, training program, and manufacturing workflow.
- Proposed and initiated overhauled of workshop by custom building 13 workbenches, 2 CNC enclosures, and 2 parts carts.

PROJECTS

ENGIN 100 - Electronics for the Internet of Things - Automatic Rice Cooker
- Developed an IoT rice cooking system using Onshape and Creaform 3D scanning to design custom mechanical interfaces
- Integrated ESP32 microcontrollers with ultrasonic and load cell sensors to automate rice portioning, washing, and cooking

ENGIN 29 - Manufacturing and Design Communication - Bike Trailer (Best Design & Integration Award):
- Designed and manufactured a collapsible bike trailer, reducing to ¼ of its original footprint for storage.
- Performed static load simulations in ANSYS to validate the frame and hinge design prior to fabrication, supporting 200+ lbs.

Combat Robotics:
- Designed, harnessed, and fabricated multiple fully 3D-printed competitive 1lb robots for local competitions.
- Modeled & prototyped a non-traditional offset slotted-link shuffler driverbase mechanism utilizing a crank and slider.

SKILLS

Onshape, Solidworks, Fusion 360, ANSYS, LSDYNA, MATLAB, Python, 3D Printing, CNC, Laser Cutting, Lathe, Mill, TIG Welding